

Fundamentals of Information Security

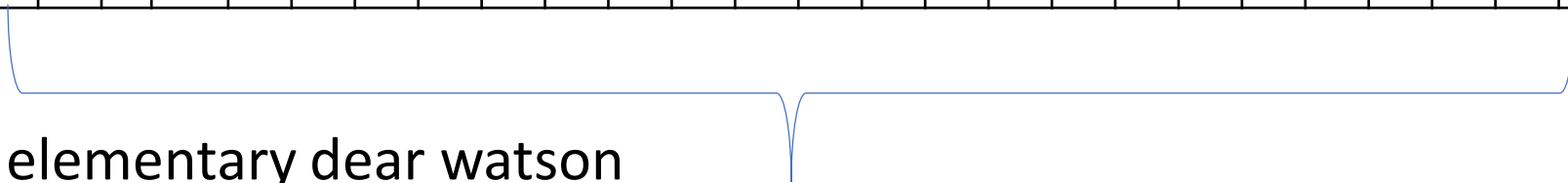
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Substitution Cipher

- Algorithm

- Substitute each letter in the plaintext by another letter specified by the key

m	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
k	G	Q	B	T	X	U	L	C	H	S	R	W	V	Z	D	E	Y	O	K	A	M	N	J	I	F	P



- Plaintext : elementary dear watson

Ciphertext : xwxvxzagof txgo jgakdz

- Here, one mapping is one key itself !!!

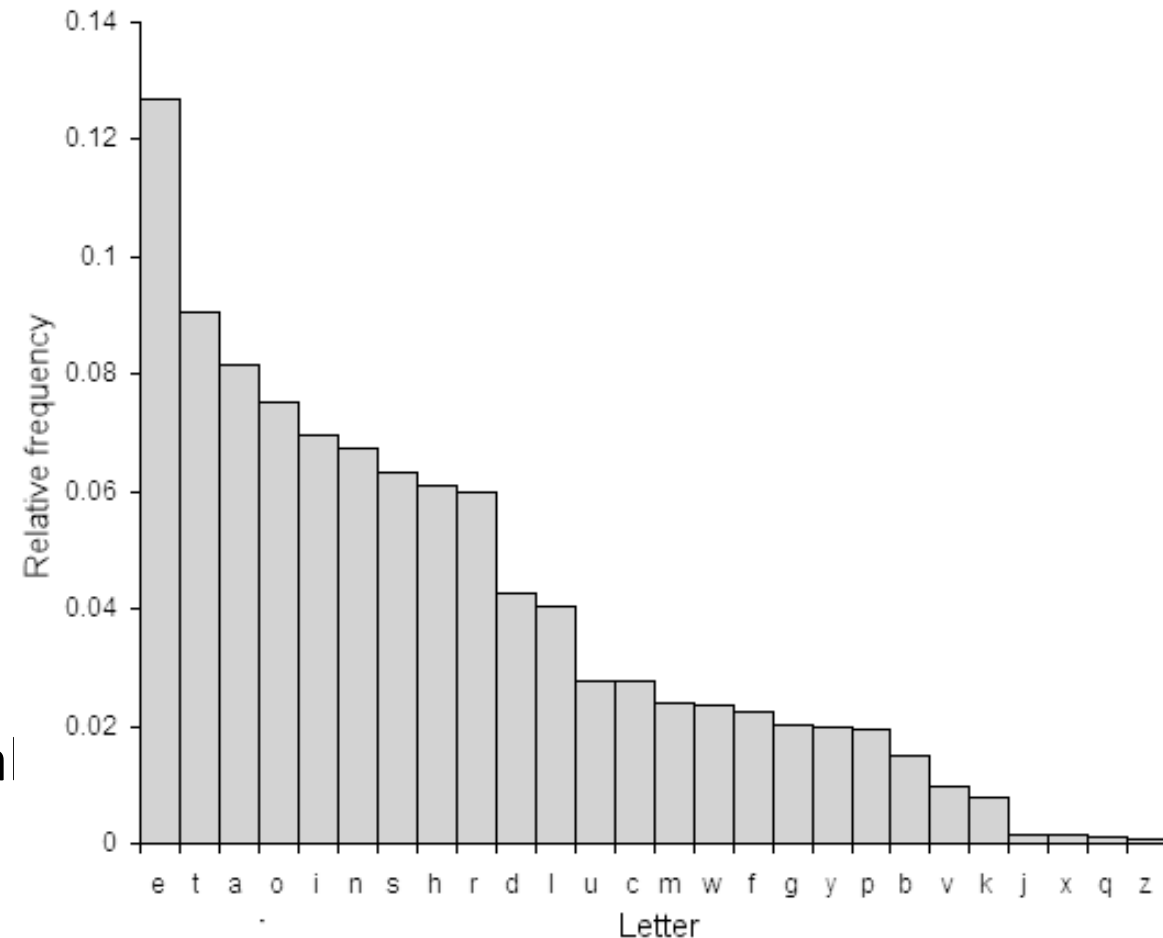
Substitution Cipher

- How many key choices possible ?
- $26 \times 25 \times 24 \times 23 \times \dots = 26!$
- $26! = 2^{88}$ approx. 10^{26} computations
- Practically impossible to do BRUTE FORCE ATTACK !!!
- So what does it mean ... substitution cipher is unbreakable ??
- Unfortunately **NO** ...

Key Size (bits)	Number of Alternative Keys	Time required at 1 decryption/ μ s	Time required at 10^6 decryptions/ μ s
32	$2^{32} = 4.3 \times 10^9$	$2^{31} \mu\text{s} = 35.8$ minutes	2.15 milliseconds
56	$2^{56} = 7.2 \times 10^{16}$	$2^{55} \mu\text{s} = 1142$ years	10.01 hours
128	$2^{128} = 3.4 \times 10^{38}$	$2^{127} \mu\text{s} = 5.4 \times 10^{24}$ years	5.4×10^{18} years
168	$2^{168} = 3.7 \times 10^{50}$	$2^{167} \mu\text{s} = 5.9 \times 10^{36}$ years	5.9×10^{30} years
26 characters permutation	$26! = 4 \times 10^{26}$	$2 \times 10^{26} \mu\text{s} = 6.4 \times 10^{12}$ years	6.4×10^6 years

Substitution Cipher

- *Frequency Analysis Technique*
- Cipher is a **monoalphabetic** cipher
 - Each plaintext letter is always going to be replaced by the same ciphertext letter
 - Frequency of letters in plaintext is preserved in the ciphertext
 - E.g., if E(plaintext) $\rightarrow$ D (ciphertext), then D will be the most frequently occurring letter in the ciphertext
 - Analyze these mappings to find potential substitutions



Substitution Cipher

- Frequency Analysis Technique

Letters								
e	t	a	o	i	n	s	r	h
12.49%	9.28%	8.04%	7.64%	7.57%	7.23%	6.51%	6.28%	5.05%

Bigrams											
th	he	in	er	an	re	on	at	en	nd	ti	es
3.56%	3.08%	2.43%	2.05%	1.99%	1.85%	1.76%	1.49%	1.45%	1.35%	1.34%	1.34%

Double Letters										
ll	ss	ee	oo	tt	ff	pp	rr	mm	cc	nn
0.58%	0.41%	0.38%	0.21%	0.17%	0.15%	0.14%	0.12%	0.10%	0.08%	0.07%

Frequently occurring 2-letter words	th, he, er, on, an, of, to, in, it, is, be, as, at, so, we, he, by, or, do, if, me, my, up, an, go, no, us, am
Frequently occurring 3-letter words	the, and, for, are, but, not, you, all, any, can, had, her, was, one, our, out, day, get, has, him, his, how, man, new, now, old, see, two, way, who, boy, did, its, let, put, say, she, too, use
Frequently occurring 4-Letter Words	that, with, have, this, will, your, from, they, know, want, been, good, much, some, time

Substitution Cipher

- **Frequency Analysis Technique**
- Correlate most frequently observed letter in ciphertext with most frequent letter in english language , then second most frequent and so on ...

LIVITCSWPIYVEWHEVSRIQMXLEYVEOIEWHRXEXIPFE
MVEWHKVSTYLXZIXLIKIIXPIJVSZEYPERRGERIMWQL
MGLMXQERIWGPSRIHMXQEREKIETXMJTPRGEVEKEITR
EWHEXXLEXXMZITWAWSQWXSWEEXTVEPMRXRSJGSTVRI
EYVIEXCVMUIMWERGMIWXMJMGCSMWXSJOMIQXLIVIQ
IVIXQSVSTWHKPEGARCSXRWIEVSWIIBXVIZMXFSJXL
IKEGAEWHEPSWYSWIWIEVXLISXLIVXLIRGEPIRQIVI
IBGIIHMWYPFLEVHEWHYPSRRFQMXLEPPXLIIECCIEVE
WGISJKTVWMRLIHYSPhXLIQIMYLSJXLIMWRIGXQER
OIVFVIZEVAEKPIEWHXEAMWYEPPXLMWYRMWXSGSWRM
HIVEXMSWMGSTPHLEVHPFKPEZINTCMXIVJSVLMRSCM
WMSWVIRCIGXMWYMX

Frequency Analysis of Ciphertext:

I – Most common letter

XL – Most common bigram

XLI – Most common trigram

E – Third most common letter

By Frequency Mapping:

I → E

XL → TH

XLI → THE

E → A

Substitution Cipher

- Frequency Analysis Technique

- Correlate most frequently observed letter in ciphertext with most frequent letter in english language , then second most frequent and so on ...

heVeTCSWPeYVaWHaVSReQMthaYVaOeaWHRtatePF
aMVaWHKVSTYhtZe~~the~~KeetPeJVSZaYPaRRGaReM
WQhMGhMtQaReWGPSReHMTQaRaKeaTtMJTPRGaVaK
aeTRaWHatthattMZeTWAWSQWtSWatTVaPMRtRSJ
GSTVReaYVeatCVMUeMWaRGMeWtMJMGCSMWtSJOMe
Q~~the~~VeQeVetQSVSTWHKPaGARCStRWeaVSWeeBtV
eZMtFSJ~~the~~KaGAaWHaPSWYSWeWeaVtheStheVthe
RGaPeRQeVeeBGeeHMYYPFhaVHaWHYPSRRFQMtha
PPtheaCCeaVaWGeSJKTVMRheHYSPHtheQeMYhtS
JtheMWRGtQaROeVFVeZaVAaKPeaWHtaAMWYaPP
thMWYRMWtSGSWRMHeVatMSWMGSTPHhaVHPFKPaZe
NTCMteVJSVhMRSCMWSWVerCeGtMWYMt

Other Possibilities:

heVe → here

Rtate → State

atthattMZe → at that time

By Frequency Mapping:

I → E

V -> R

XL → TH

R -> S

XLI → THE

M -> I

E → A

Z -> M

Substitution Cipher

- *Frequency Analysis Technique*
- Only 1 ciphertext is enough to get the key
 - But the ciphertext should be long enough (atleast 30 characters) to make statistical correlation

- The remaining concepts will be discussed and explained in class...!